

**LINDA CHICK.**  
**Automated Poultry Coop System.**  
**Executive Summary**

The Automated Poultry Coop System is an integrated hardware–software IoT solution designed to modernize small-scale and medium-scale poultry farming. It combines an automated physical coop, a robotic feeding and monitoring mechanism (Slider Chick), and a mobile application (ChickPal) connected through a real-time MQTT communication system.

The system reduces manual labor, improves chick survival rates, optimizes feeding efficiency, and enables farmers to monitor and control their poultry operations remotely. By addressing inefficiencies, losses, and poor monitoring in traditional poultry farming, the solution increases productivity, profitability, and sustainability.

**Problem Statement**

Poultry farming is one of the most common and accessible income sources, especially for smallholder and youth farmers. However, most poultry operations still rely on manual processes that are inefficient, time-consuming, and error-prone.

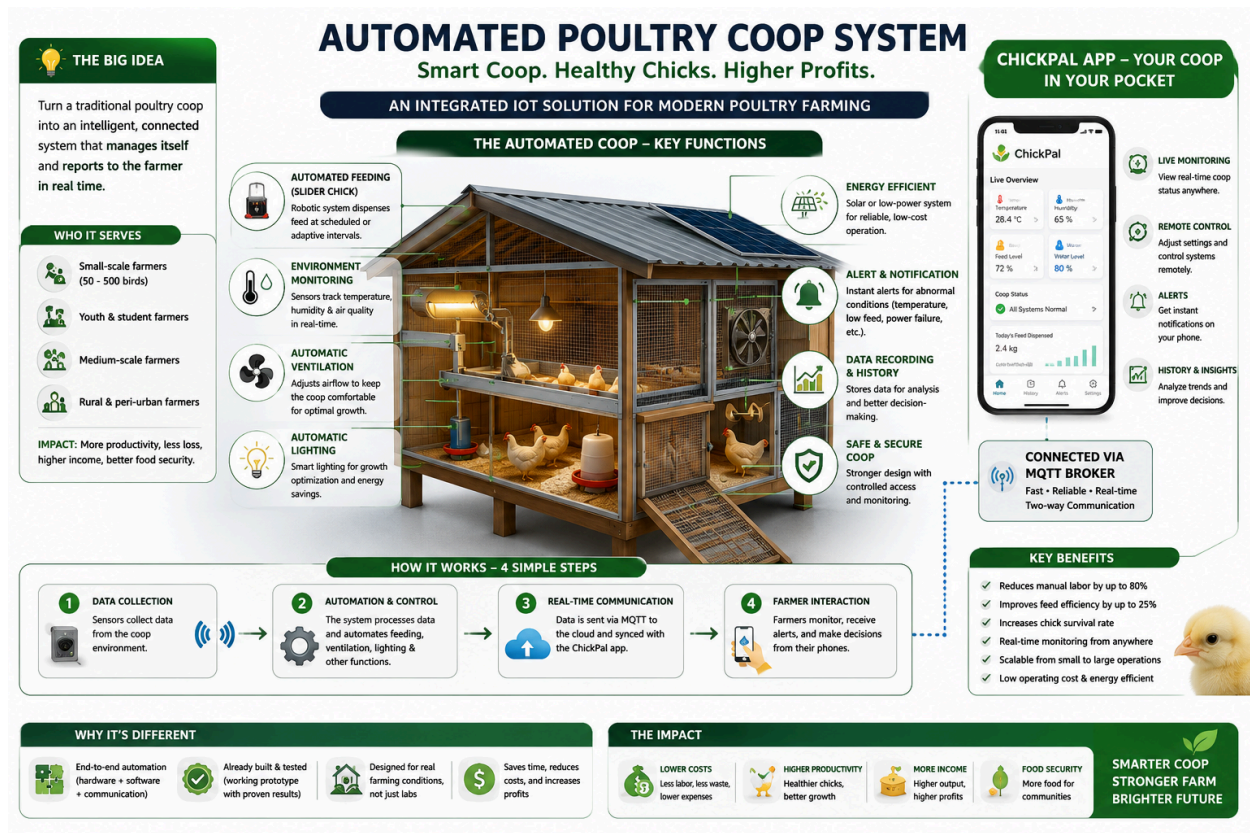
Key challenges include:

- Irregular feeding schedules leading to poor growth
- Poor environmental control (temperature, humidity, ventilation)
- High chick mortality due to delayed detection of issues
- Lack of real-time monitoring when farmers are away
- Labor-intensive daily routines that limit scalability

These problems result in:

- Financial losses
- Reduced productivity
- Farmer burnout
- Limited ability to scale poultry businesses

This problem matters because improving poultry efficiency directly improves food security, income stability, and economic resilience at the household and community level.



## Who Is Affected .

The problem primarily affects:

- **Small-scale poultry farmers** with 50–500 birds
- **Youth and student farmers** balancing farming with school or work
- **Medium-scale farmers** seeking to scale without hiring more labor
- **Rural and peri-urban farmers** with limited access to advanced farm technology

Indirectly affected:

- Consumers (food availability and pricing)
- Local economies dependent on poultry value chains.

# The Big Idea

The core idea is simple but powerful:

**Turn a traditional poultry coop into an intelligent, connected system that manages itself and reports to the farmer in real time.**

Instead of farmers constantly checking the coop, the coop checks itself — and reports to the farmer through an app.

## 5. Solution Overview

The Automated Poultry Coop System solves these challenges by integrating:

1. **Automated Coop Hardware**  
Controls feeding, environment, and movement with minimal human intervention.
2. **Slider Chick Mechanism**  
A robotic or guided feeding and monitoring system that distributes feed efficiently and consistently.
3. **ChickPal Mobile App**  
Allows farmers to monitor, control, and receive alerts from anywhere.
4. **MQTT Communication Broker**  
Enables fast, reliable, real-time data exchange between hardware and software components

### Primary Objective

Develop an affordable automated poultry management system that improves efficiency and reduces manual labor.

### Secondary Objectives

- Improve chick survival.
- Reduce feeding inconsistencies.
- Enable remote monitoring.

- Improve environmental control.
- Increase farm productivity.
- Support scalable poultry businesses.

## **How the System Works (Implementation)**

### **Step 1: Data Collection**

Sensors inside the coop measure:

- Temperature
- Humidity
- Feed levels
- Activity or movement patterns

### **Step 2: Automation & Control**

- Slider Chick dispenses feed at scheduled or adaptive intervals
- Environmental conditions are adjusted automatically
- Abnormal conditions trigger alerts

### **Step 3: Real-Time Communication**

- Data is transmitted via MQTT to the backend
- The ChickPal app displays live status and notifications

### **Step 4: Farmer Interaction**

- Farmers view coop status on their phone
- Manual overrides are possible when needed
- Historical data supports better decision-making

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## **User Stories**

**Farmer**

"I want to know the temperature of my coop while away from home."

System

Displays live temperature.

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**Farmer**

"I want feeding to happen automatically."

System

Slider Chick dispenses feed according to schedule.

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**Farmer**

"I want to receive alerts on how my chicks are doing."

System

Send push notification immediately.

## **Key Features of the Automated Coop System**

- Automated feeding system (Slider Chick)
  - Real-time environmental monitoring
  - Remote monitoring via ChickPal app
  - MQTT-based real-time communication
  - Alert and notification system
  - Modular hardware design
  - Energy-efficient operation
  - Scalable for different coop sizes
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## Cost Structure

### Estimated Costs (Prototype / Early Deployment)

- Sensors and microcontrollers
- Mechanical feeding system
- Structural coop modifications
- Power supply (electric or solar)
- App development and backend services

Costs are designed to be **modular**, allowing farmers to start small and upgrade gradually.

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## Revenue Model

Potential revenue streams include:

- Sale of automated coop kits
  - Subscription fees for the ChickPal app (premium features)
  - Maintenance and upgrade services
  - Licensing to poultry cooperatives or institutions
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## Is the System Self-Sustaining?

Yes.

Once installed:

- Automation reduces daily labor costs
- Improved feed efficiency lowers operational expenses
- Increased survival rates boost revenue
- App subscriptions fund ongoing software maintenance

This creates a sustainable ecosystem where the system pays for itself over time.

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## Investor Value & Returns

Investors gain:

- Entry into the growing Agri-IoT market
- A product with proven traction (WRO exposure and working prototype)
- A scalable hardware-software business model
- Opportunities for expansion into data analytics and farm intelligence

Returns grow as:

- More units are deployed
  - Subscription base increases
  - Data-driven services are added
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## Resources Required

### Functional requirements

- Feed dispensing automation
- Environmental monitoring
- Real-time alerts
- Remote control via app

### Non-functional requirements

- Reliability (24 hours 7 days a week)
- Low latency (MQTT real-time)
- Security (device authentication)
- Offline tolerance (local fallback logic)

### **1. Hardware Components.**

The system relies on robust, affordable, and locally serviceable hardware components, including:

- Microcontrollers (for example ESP32)
- Environmental sensors (temperature, humidity, gas levels)

- Actuators and motors for the Slider Chick feeding system
- Power systems (grid and/or solar-supported)
- Coop structural adaptations and enclosures

Manufacturing and assembly partners are required to:

- Produce modular hardware kits
- Ensure quality control and durability
- Support localized assembly to reduce costs
- Enable scalable production as demand grows

Strategic partnerships with local manufacturers reduce logistics costs and improve maintainability in rural settings.

## Why This Solution Is Different

Unlike existing solutions:

- It integrates **hardware + software + real-time communication**
- It is **already built and tested**, not just conceptual
- It is designed for **practical farming realities**, not ideal labs
- It reduces labor instead of adding complexity

Most competitors focus on isolated tools; this system delivers **end-to-end automation**.

## Conclusion

The Automated Poultry Coop System transforms poultry farming from a manual, labor-intensive activity into a smart, efficient, and scalable operation. By combining automation, IoT, and real-time monitoring, it empowers farmers to increase productivity, reduce losses, and grow sustainably.